Correcting for flip angle inhomogenuities in quantitative MRI T2 mapping

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Abstract

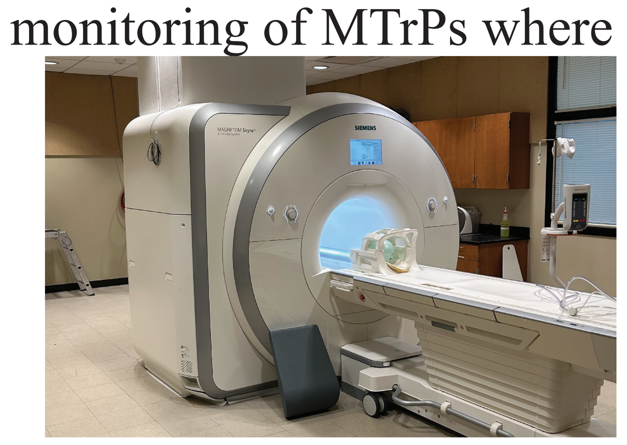
In magnetic resonance imaging, T2 mapping is a technique that is used to quantitatively map individual voxels to T2 relaxation times by taking signal measurements at a series of different echo times and fitting them to a decay curve. T2 maps have a wide range of research applications in different tissues, generally indicating the water content of a tissue, which can then be used to make some statement about the health and other properties of the region, independent of anatomical structure. While these images can be invaluable resources in a research setting, problems such as fitting error artifacts begin to arise due to variation in the generated pulse sequence flip angle, especially in higher field strength systems, due to the increase in T2 times, demanding more data points to accurately model. In order to take full advantage of the higher resolutions that high-field systems offer, we must correct for the variation in flip angle as it is amplified by the higher B0 fields used in research settings. While all the methods explored can be tuned to produce realistic results, the extended phase graph method proved strongest, generating T2 times consistent with literature while also accounting for flip angle variation that other methods fail to address. A pipeline was developed enabling the research community to process large multi-patient datasets, often in less than an hour.

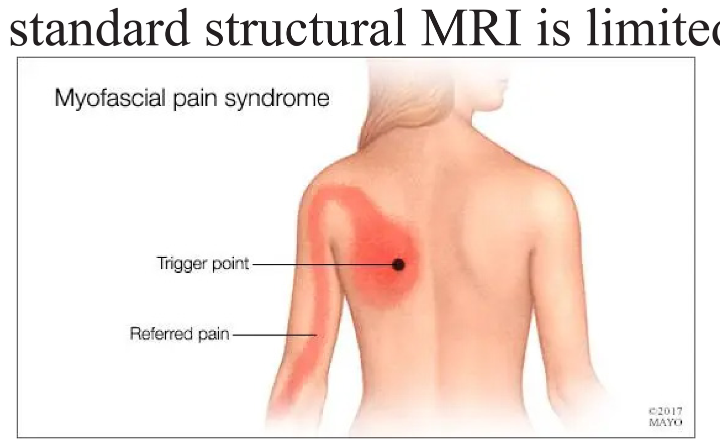
Background

In MRI and nuclear magnetic resonance in general, T2 time can be characterized as the time it takes for a molecule's pulse-induced transverse magnetization (Mxy) to decay to 37% of its value immediately after the pulse (M0). This relationship can be described with the simplified Bloch equation shown

$$M_{xy}(TE) = M_0 * e^{-\frac{TE}{T2}}$$

As a direct clinical application, we are exploring whether quantitative T2 mapping may address the diagnostic challenges of Myofascial Pain Syndrome (MPS). The lack of an objective tool to identify its myofascial trigger points (MTrPs) presents a critical need. We hypothesize that our methods can provide a quantitatively robust approach to the identification and monitoring of MTrPs where standard structural MRI is limited.





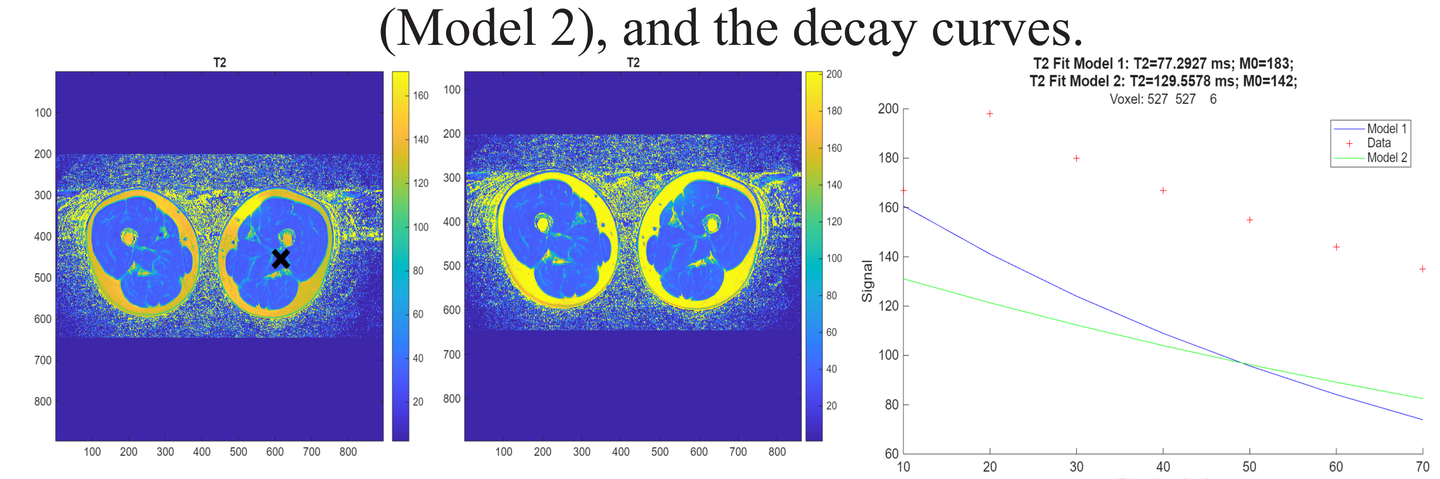
Methods

- <u>Participants:</u>
  - Initial survey of tooling conducted using a single calf scan to simplify anatomical segmentation.
  - Findings were confirmed over a data set of 56 patients and 112 scans from MTrPs study of trapezius muscle
- <u>Imaging:</u>
  - All scans were completed on a single Siemens MAGNETOM Skyra Fit 3T system using a 32 channel spinal coil
  - 11 slices were obtained at 896x896
  - Mxy was measured at 7 echo times spaced 10ms apart for each voxel
  - Siemens commerical multi-contrast spin echo sequence (named 'se_mc' in software) was used to implement the Carr-Purcell-Meiboom-Gill NMR sequence
- <u>T2 Mapping:</u>
  - qMRLab mono-exponential fit
    - Both with and without first echo measured T2
  - Decaes.jl multi-exponential extended phase graph T2 mapping toolbox

Monoexponential Fitting

Monoexponential fitting is a basic method of generating T2 maps. The process consists of taking a given voxel's value for each echo as a point in time and fitting the simplified Bloch in order to quantitatively measure the T2 time.

With this method, it is important to drop the first echo time in order to produce acceptable results. This is because the simplified Bloch equation models the signal from a uniform sample subject to a perfect pulse sequence, but in MRI, the signal we measure is subject to interference from a multitude of factors, such as neighboring voxels, field inhomogeneities, and flip angle variations, to name a few. Dropping the first echo time can reduce artifacts in the mono-exponential fit and is recommended by Siemens for this sequence. Below in order, T2 maps without the first echo (Model 1), with the first echo (Model 2), and the decay curves.

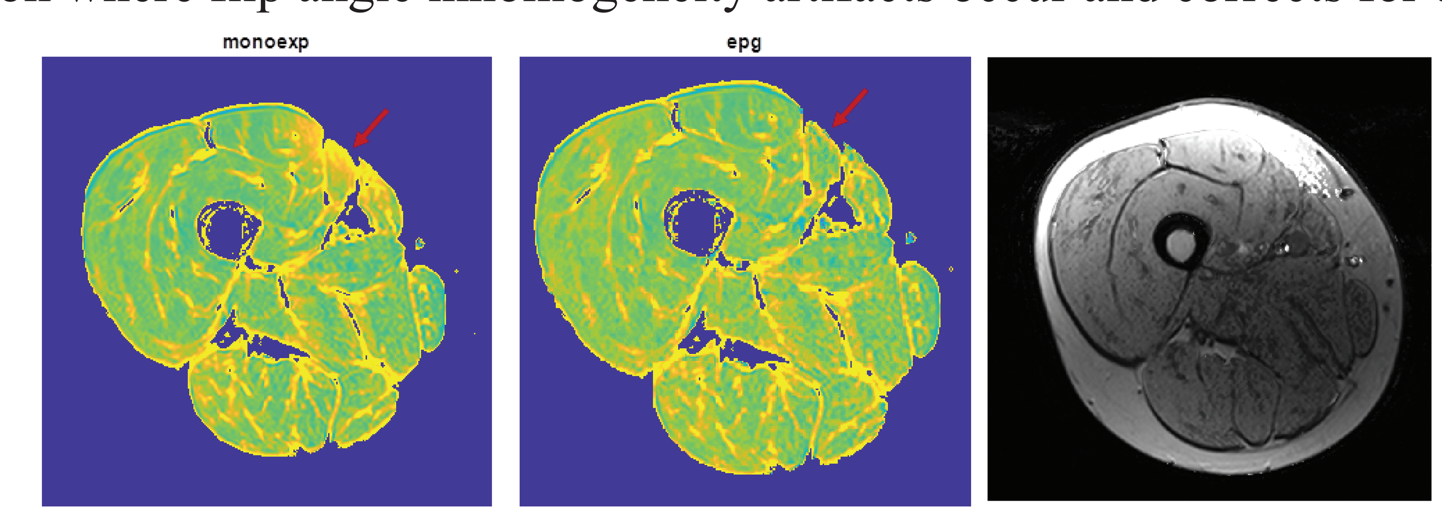


Multi-Exponential Fitting

In order to address the artifacting issues encountered with mono-exponential fitting, we must account for the additional components of our measurements that the simplified Bloch equations fail to acknowledge. While many unwanted components contribute to our data, correcting for refocusing pulse flip angle variations can often remove a large portion of distortions. When we run a CPMG sequence, we expect the refocusing pulse to have a flip angle of 180 degrees; however, this is often not the case due to limitations of the scanner. In order to correct for this, we employed the extended phase graph method to construct a matrix of exponential decay bases (A), which are then used to perform non-negative least squares fitting, yielding a spectrum of exponential decay component amplitudes. The primary decomposition method is founded on an inverse Laplace transform technique, which requires solving the regularized nonnegative least squares (NNLS) inverse problem shown below.

$$X_\mu = \underset{x \geq 0}{\arg\min} \|Ax - b\|_2^2 + \mu^2 \|x\|_2^2$$

This method handles unwanted signal components in a much more robust way than the mono-exponential fit does. As we can see below in the total T2 signal density map shown in grayscale, this approach correctly identifies the region where flip angle inhomogeneity artifacts occur and corrects for them.



Results

After both qualitative and basic quantitative comparison of the techniques explored, we can comfortably conclude that the extended phase graph method is the most effective approach to reduce T2 artifacting. Subsequently, a MATLAB pipeline was constructed to enable simple T2 mapping and visual comparison of methods, both voxel-wise and wholistically.

References

- J. Doucette, C. Kames, and A. Rauscher, "DECAES – DEcomposition and Component Analysis of Exponential Signals," Zeitschrift für Medizinische Physik, vol. 30, no. 4, pp. 271–278, Nov. 2020, doi: 10.1016/j.zemedi.2020.04.001.
- A. Karakuzu et al., "qMRLab: Quantitative MRI analysis, under one umbrella," JOSS, vol. 5, no. 53, p. 2343, Sept. 2020, doi: 10.21105/joss.02343.a

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